Online PAC Labeling with Joint Error and Cost Guarantees

Selective expert feedback, constrained routing, and a matching cost exponent

Abstract

We study online labeling with a cheap predictor and a costly, correct expert. Each expert query both prevents a released-label error and reveals feedback about the predictor; released model labels supply no feedback. The objective is to control cumulative released error while approaching the cost of an optimal feasible score-based policy. Under i.i.d. arrivals, a fixed predictor and score, and a known global Lipschitz bound on conditional model error, we propose one-sided barrier saddle routing (OBSR). It learns a possibly nonmonotone release policy and an error price from selectively revealed labels. We prove simultaneous high-probability bounds of order $\widetilde{O}(T^{2/3})$ on cumulative error excess and cost regret against all feasible Borel score policies. An explicit target-tightening condition gives nominal PAC error control with sublinear excess cost. A complementary information-theoretic construction yields a $T^{2/3}$ lower bound on a weighted cost–error tradeoff. Together, these results determine the minimax cost exponent, up to logarithmic factors, for a specified nominal-PAC class with failure probability T^{-2}. We provide explicit updates and complete proofs; empirical validation is left to future work.

1 From error control to cost-efficient labeling

Large-scale labeling combines inexpensive model predictions with costly expert judgments. A useful guarantee must address both the accuracy of the released labels and the cost of producing them. Sending every item to a correct expert trivially controls error; achieving that control while learning when the model can safely be used is the statistical problem. We ask whether an online learner can approach the cost of an optimal error-constrained policy when true labels are available only through paid expert queries.

The optimization principle. Let S be an observed score and $m(s)$ the model's conditional error probability. A score policy releases the model label with probability $a(s)$ and otherwise uses the expert. With unit expert cost, its expected expert expenditure and released error are $\mathbb{E}[1 - a(S)]$ and $\mathbb{E}[a(S)m(S)]$. The population oracle solves

$$\inf_{a:[0,1]\to[0,1]} \mathbb{E}[1 - a(S)] \quad \text{subject to} \quad \mathbb{E}[a(S)m(S)] \le \varepsilon, \tag{1}$$

where policies are Borel measurable. At a fixed error price $\lambda \ge 0$, the Lagrangian integrand is $1 + a(s)(\lambda m(s) - 1)$. It favors release when $\lambda m(s) < 1$ and deferral when $\lambda m(s) > 1$, with randomization possible at equality. This linear-programming interpretation explains the role of the price; it is a reformulation, not a novelty claim. The learner must estimate useful routing decisions without knowing m or receiving unpaid labels. Moreover, a threshold in $m(s)$ need not be a threshold in s: the score need not be calibrated or monotone in actual error.

Method and results. OBSR learns release probabilities on score bins together with a dual error price. Its one-sided log barrier accommodates the inverse-probability feedback supplied by expert deferrals. A vanishing query floor maintains information while limiting exploration cost. The main results are a simultaneous finite-horizon error and cost guarantee (Theorem 4.1), a sufficient condition for nominal error control (Corollary 4.3), and a lower bound for every adaptive learner in a specialized instance family (Theorem 6.1). Corollary 6.3 connects the upper and lower results under a precisely defined nominal-PAC criterion. The comparator knows the population law but cannot inspect the current true label before deciding. All paid model and expert calls are counted.

Relation to existing methods. PAC labeling provides certification for a fixed finite pool, including multiple-model routing, learned uncertainty scores, and cost-sensitive variants [1]. Its guarantees do not require an i.i.d. population or a smooth score–error relation. Our problem instead concerns sequential learning from paid feedback and cost regret against a population oracle. Learning to defer also optimizes the allocation of predictions between a model and an expert; consistent surrogate methods learn from labeled training data and can allow an imperfect expert [2]. Their statistical target differs from a finite-horizon released-error constraint with online acquisition costs.

Action-dependent label revelation is central to apple tasting and partial monitoring [3]. Online optimization with stochastic constraints studies simultaneous objective regret and constraint violation [4], but its full-feedback formulation supplies constraint functions after each decision. Here that information must be estimated from the labels actually purchased. Log-barrier mirror descent [5] and adaptive squared-gradient scaling [6] are established ingredients. Our results concern their combination with score approximation, selective feedback, and a common cost–error comparator; they do not claim these ingredients or cost-sensitive routing as new ideas. Section 4.2 makes the comparison of information and guarantees explicit.

Organization and prerequisites. Section 2 defines the information protocol, assumptions, and comparator. Section 3 gives the complete algorithm, and Sections 4–5 state and prove its guarantees. Section 6 gives the lower bound and minimax consequence, with the full construction in Appendix A. The proofs use conditional expectation, convex duality, and elementary concentration and information inequalities. The mirror-descent estimate and concentration bound needed for the upper result are proved below. We use $\widetilde{O}$ to suppress logarithmic factors; exact remainders and their parameter dependence are displayed in the theorems.

2 Stochastic model and population benchmark

Assumption 1 (Fixed stochastic source)**.** *The triples* $(S_t, \widehat{Y}_t, Y_t)_{t=1}^T$ *are i.i.d., with score* $S_t \in [0,1]$ *and zero–one model loss* $L_t = \mathbf{1}\{\widehat{Y}_t \neq Y_t\}$*. The model and score mechanism are fixed before the stream. No expert labels outside the specified feedback protocol are available to the learner.*

Assumption 2 (Costs and safe expert)**.** *Calling the model costs a known constant* $c_M \in [0,1]$ *on every round. An expert query costs one and returns the correct label* Y_t*. An audit-and-correct action has the same expert charge and an additional known cost* $c_R \in [0,1]$ *when the model label is wrong.*

Assumption 3 (Global score regularity)**.** *The conditional error* $m(s) = \mathbb{P}(L_t = 1 \mid S_t = s)$ *admits a version on* $[0,1]$ *with a known finite bound* $L \geq 0$ *such that*

$$|m(s) - m(s')| \leq L|s - s'|, \qquad s, s' \in [0,1]. \tag{2}$$

These assumptions describe a stationary labeling population and a reliable expert. They impose neither calibration $m(s) = s$, monotonicity of m, a score-density lower bound, nor conditions near an unknown optimal threshold. Distribution shift, online retraining of the source, and unrestricted changes to the score mechanism fall outside the theorem. Observed drift, expert disagreement, or large local error differences can challenge the assumptions; finite data alone do not certify a global Lipschitz bound without additional structure.

2.1 Information, losses, and costs

At round t, the learner first fixes its state from the past, pays c_M, and observes $(S_t, \widehat{Y}_t)$. It then chooses release, defer, or audit using observable information and fresh randomness independent of the current label. On release it outputs $\widehat{Y}_t$ and never observes Y_t. Both expert actions reveal Y_t and output that label; an audited error is corrected. Only then is the state updated. Let $\mathcal{H}_{t-1}$ denote the observable history and state before the current triple arrives. All initial-state and update kernels are fixed independently of the unknown distribution.

For a general learner let $A_t \in \{\mathsf{release}, \mathsf{defer}, \mathsf{audit}\}$ and $I_t = \mathbf{1}\{A_t \neq \mathsf{release}\}$. Its actual released loss and total cost are

$$R_t = (1 - I_t)L_t, \qquad \mathcal{R}_T = \sum_t R_t, \qquad C_T = c_M T + \sum_t (I_t + c_R \mathbf{1}\{A_t = \mathsf{audit}\} L_t). \tag{3}$$

The latent counterfactual loss L_t, the released loss R_t, and any feedback estimate are different objects. In particular, a corrected label has $R_t = 0$. The proposed algorithm routes using the score and history; the model prediction is used as the candidate output, not as an additional routing feature.

2.2 The comparator

For a Borel release probability $a : [0,1] \to [0,1]$, define

$$r_P(a) = \mathbb{E}[a(S)m(S)], \qquad c_P(a) = \mathbb{E}[1 - a(S)], \qquad d_0(\varepsilon) = \inf_{a:\, r_P(a) \leq \varepsilon} c_P(a), \tag{4}$$

where the infimum is over such Borel functions. For $\varepsilon \in (0,1]$, the benchmark is

$$C_T^\star(\varepsilon) = T\{c_M + d_0(\varepsilon)\}. \tag{5}$$

This policy may know the distribution, but cannot see the current true label before acting. It uses the same score and pays the same model and expert costs. Auditing is weakly dominated by direct deferral in this model: both actions reveal and release Y_t, but audit corrections can cost extra. Thus allowing audits does not lower this benchmark. Both learner and comparator call the model on every round; a policy that can skip this call is a different comparison problem.

We seek a simultaneous fixed-horizon event on which

$$\mathcal{R}_T \leq \varepsilon T + o(T), \qquad C_T \leq C_T^\star(\varepsilon) + o(T). \tag{6}$$

The benchmark cost is expected population cost, while C_T and $\mathcal{R}_T$ are realized quantities. Probabilities in the upper bound include both the i.i.d. data and the learner's coins.

3 One-sided barrier saddle routing

Partition $[0,1]$ into K equal bins, assigning a boundary to the bin on its right and including 1 in the last bin. Write $b(s)$ for its bin. For an integer horizon $T \geq 2$, choose $K \geq 1$ and $\gamma \in [1/T, 1)$. Maintain release probabilities $x_t \in \mathcal{X}_\gamma = [0, 1-\gamma]^K$ and a price $\lambda_t \in [0, \Lambda]$, where

$$\Lambda = \frac{2}{\varepsilon}, \qquad \bar{\varepsilon} = \min\{1, \varepsilon + L/K\}, \qquad \eta_x = \sqrt{\frac{K \log(1/\gamma)}{2(1+\Lambda^2)T}}. \tag{7}$$

The relaxed internal budget $\bar{\varepsilon}$ pays for score discretization; it is accounted for explicitly in the final error bound. The reveal floor γ is an algorithm choice, not an assumption on the population.

Algorithm 1 OBSR: one-sided barrier saddle routing

Require: $T, \varepsilon, L, K, \gamma$; parameters from (7).

Initialize $x_1 = 0 \in \mathbb{R}^K$, $\lambda_1 = 0$, $B_0 = 0$.
for $t = 1, \ldots, T$ **do**
  Pay c_M and observe $(S_t, \widehat{Y}_t)$; set $b_t = b(S_t)$.
  Set $z_t = x_{t,b_t}$ and $q_t = 1 - z_t$; draw $D_t \sim \mathrm{Ber}(q_t)$.
  if $D_t = 1$ **then**
    Pay one, observe Y_t, output Y_t, set $W_t = \mathbf{1}\{\widehat{Y}_t \neq Y_t\}/q_t$.
  else
    Output $\widehat{Y}_t$, observe no true label, and set $W_t = 0$.
  end if
  Form $g_{t,b_t} = -1 + \lambda_t W_t$ and $h_t = z_t W_t - \bar{\varepsilon}$.
  Update coordinate b_t by (12); leave all other coordinates unchanged.
  Set $B_t = B_{t-1} + h_t^2$ and update λ_{t+1} by (13).
end for

At score S_t, let $z_t = x_{t,b_t}$, $b_t = b(S_t)$, and $q_t = 1 - z_t$. Draw $D_t \sim \mathrm{Ber}(q_t)$ independently of the latent label conditional on the past and current observation. If $D_t = 1$, defer to the expert; otherwise release the model label. Form the computable estimator

$$W_t = \begin{cases} L_t/q_t, & D_t = 1, \\ 0, & D_t = 0, \end{cases} \qquad g_t = e_{b_t}(-1 + \lambda_t W_t), \qquad h_t = z_t W_t - \bar{\varepsilon}. \tag{8}$$

Here e_j is the jth standard basis vector of $\mathbb{R}^K$. Conditional on the past and score, $\mathbb{E}[W_t \mid \mathcal{H}_{t-1}, S_t] = m(S_t)$. Setting $W_t = 0$ after release is computable and does not assert that the hidden error is zero.

The population saddle objective on the bins is

$$d(x) + \lambda(r(x) - \bar{\varepsilon}), \qquad d(x) = \mathbb{E}[1 - x_{b(S)}], \quad r(x) = \mathbb{E}[x_{b(S)} m(S)]. \tag{9}$$

The vectors in (8) estimate its primal and dual gradients. For the primal update use the one-sided log barrier and its Bregman divergence,

$$\Psi(x) = -\sum_{j=1}^{K} \log(1 - x_j), \qquad D_\Psi(u, v) = \Psi(u) - \Psi(v) - \langle \nabla\Psi(v), u - v\rangle. \tag{10}$$

One mirror-descent step is

$$x_{t+1} = \arg\min_{x \in \mathcal{X}_\gamma} \{\eta_x \langle g_t, x\rangle + D_\Psi(x, x_t)\}. \tag{11}$$

It changes only coordinate b_t. With $\mathrm{clip}_{[a,b]}(v) = \min\{b, \max\{a, v\}\}$, its exact solution is

$$v_t = q_t^{-1} - \eta_x g_{t,b_t}, \qquad x_{t+1,b_t} = 1 - \frac{1}{\mathrm{clip}_{[1,1/\gamma]}(v_t)}. \tag{12}$$

Indeed the derivative of the scalar objective is $1/(1-x) - v_t$. This formula handles boundary and large-step cases without an optimization solver. The dual update is projected ascent with adaptive scaling:

$$B_t = \sum_{u=1}^{t} h_u^2, \qquad \eta_{\lambda,t} = \frac{\Lambda}{\sqrt{1 + B_t}}, \qquad \lambda_{t+1} = \mathrm{clip}_{[0,\Lambda]}(\lambda_t + \eta_{\lambda,t} h_t). \tag{13}$$

OBSR uses no separate audit action: every acquired label is an expert deferral, so $C_T = c_M T + \sum_t D_t$ and $R_t = (1 - D_t)L_t$. Deferral supplies the counterfactual model loss needed for learning while ensuring a correct output on that round. After $O(K)$ initialization, the routing updates take $O(1)$ arithmetic per round and $O(K)$ memory, excluding model and expert calls. No score monotonicity is imposed on x_t.

4 Joint guarantees and parameter choice

Theorem 4.1 (Finite-horizon OBSR guarantee)**.** *Under Assumptions 1–3, let $T \geq 2$ be an integer, $\varepsilon \in (0, 1]$, $\delta \in (0, 1)$, $K \geq 1$ an integer, and $1/T \leq \gamma < 1$. Define*

$$\begin{aligned} \ell_T &= \log(16KT/\delta), \\ H_T &= \sqrt{KT\ell_T} + \sqrt{T\ell_T/\gamma} + \ell_T/\gamma, \\ G_T(\varepsilon) &= 100(1 + \varepsilon^{-1})^2 H_T. \end{aligned} \tag{14}$$

Algorithm 1 satisfies, simultaneously with probability at least $1 - \delta$,

$$\mathcal{R}_T \leq \varepsilon T + \frac{LT}{K} + G_T(\varepsilon), \qquad C_T \leq C_T^\star(\varepsilon) + \gamma T + G_T(\varepsilon). \tag{15}$$

Corollary 4.2 (Balanced tuning)**.** *Take $K = \lceil T^{1/3} \rceil$ and $\gamma = T^{-1/3}$. On the event in Theorem 4.1,*

$$\begin{aligned} \mathcal{R}_T - \varepsilon T &\leq LT^{2/3} + \overline{G}_T, \\ C_T - C_T^\star(\varepsilon) &\leq T^{2/3} + \overline{G}_T, \\ \overline{G}_T &= 100(1 + \varepsilon^{-1})^2 \big[(1 + \sqrt{2})T^{2/3}\sqrt{\ell_T} + T^{1/3}\ell_T\big]. \end{aligned} \tag{16}$$

For fixed ε, L and fixed or polynomially small δ, both remainders are $O_{\varepsilon,L}(T^{2/3}\sqrt{\log(T/\delta)}) = o(T)$. The average released error is consequently at most $\varepsilon + \widetilde{O}(T^{-1/3})$.

Proof. Use $K \leq 2T^{1/3}$, $LT/K \leq LT^{2/3}$, and substitute into (14). The $T^{1/3}\ell_T$ term is asymptotically smaller than $T^{2/3}\sqrt{\ell_T}$ for the stated confidence levels. □

The three terms in H_T represent primal learning over K bins, importance-weighted feedback variance, and its maximal magnitude. Discretization contributes LT/K to error; the reveal floor contributes γT to cost. This makes the $T^{2/3}$ balance explicit.

4.1 Nominal PAC control by tightening the target

The bound in (15) targets ε with finite-sample slack. The following consequence explains when one can certify the requested nominal error while retaining the comparator at that requested target.

Corollary 4.3 (A sufficient condition for nominal error control)**.** *Fix a requested target $\varepsilon \in (0, 1]$ and the other parameters of Theorem 4.1. Let*

$$\widehat{G}_T = 100(1 + 2/\varepsilon)^2 H_T, \qquad b_T = \frac{L}{K} + \frac{\widehat{G}_T}{T}. \tag{17}$$

If $b_T \leq \varepsilon/2$, run OBSR with internal target $e = \varepsilon - b_T$, recomputing (7) using e. Then, simultaneously with probability at least $1 - \delta$,

$$\mathcal{R}_T \leq \varepsilon T, \qquad C_T \leq C_T^\star(\varepsilon) + \gamma T + \widehat{G}_T + \frac{b_T T}{\varepsilon}. \tag{18}$$

Under balanced tuning and fixed ε, L, the condition eventually holds for fixed or polynomially small δ, and the excess cost in (18) is $\widetilde{O}(T^{2/3})$.

Proof. Since $e \geq \varepsilon/2$, $G_T(e) \leq \widehat{G}_T$. Theorem 4.1 at target e gives $\mathcal{R}_T \leq eT + LT/K + G_T(e) \leq \varepsilon T$. For any policy feasible at ε, the scaled release probability $a_e = (e/\varepsilon)a$ is feasible at e and costs at most $c_P(a) + b_T/\varepsilon$. Taking an infimizing sequence yields $C_T^\star(e) \leq C_T^\star(\varepsilon) + b_T T/\varepsilon$. Apply the cost bound at e and combine these inequalities. □

This is a sufficient finite-sample condition, not a promise of useful savings at practical horizons. The displayed constants are conservative. When the condition fails, always using the expert still ensures zero released error, but need not have sublinear cost regret.

4.2 Comparison with public frameworks

Framework	Data and feedback	Guarantee or optimization target
PAC labels [1]	Fixed finite pool; randomized expert sampling; no score-quality assumption	High-probability control of pool-average label loss; includes cost-sensitive routing.
Learning to defer [2]	I.i.d. labeled training data with expert predictions; expert may err	Consistent surrogate estimation of the expected loss of a classifier–expert system.
Stochastic constraints [4]	Online decisions; objective and constraint functions revealed afterward; feasibility assumptions	Objective regret and cumulative constraint violation relative to an expected-feasible comparator.
Partial monitoring [3]	Sequential actions; feedback depends on action and outcome	Unconstrained loss regret against the best fixed action in hindsight.
OBSR (this work)	I.i.d. stream; known Lipschitz bound; true labels revealed only by paid, correct expert	Joint realized error and cost bounds against all feasible score policies; nominal error via target tightening.

The comparison is between mathematical formulations, not measured performance. A fixed-pool certificate and a population regret bound answer different questions, and the full-feedback stochastic-constraint theorem cannot be applied directly to selectively observed labels. Likewise, an unconstrained partial-monitoring rate does not establish the cost lower bound in Section 6. The OBSR guarantee permits nonmonotone score policies, but requires stationarity, score regularity, and a correct expert. We do not infer a uniform rate improvement over these public methods from guarantees with different assumptions.

5 Proof of the OBSR guarantee

For the analysis, let $\mathcal{F}_t$ contain all triples and learner coins through round t, including labels that were never revealed, and set $\mathbb{E}_{t-1} = \mathbb{E}[\,\cdot \mid \mathcal{F}_{t-1}]$. The observable history satisfies $\mathcal{H}_t \subseteq \mathcal{F}_t$; the learner still uses only $\mathcal{H}_{t-1}$ and the permitted current observation. The state (x_t, λ_t) is $\mathcal{F}_{t-1}$-measurable, and i.i.d. arrivals and fresh coins give the conditional means below under this enlarged filtration. In particular, $R_t - \mathbb{E}_{t-1}R_t$ is now an adapted martingale difference although an unrevealed R_t is not observable. Write $\mathcal{X}_\gamma = [0, 1-\gamma]^K$ and let

$$d_\gamma^\star = \min_{x \in \mathcal{X}_\gamma:\ r(x) \leq \bar{\varepsilon}} d(x). \tag{19}$$

An optimum exists by compactness and the safe policy $x = 0$.

Lemma 5.1 (Discretization and the safe-action multiplier)**.** *We have $d^\star_\gamma \le d_0(\varepsilon) + \gamma$. There is an optimal multiplier $0 \le \lambda^\star \le 1/\bar\varepsilon \le 1/\varepsilon$ such that*

$$d(x) - d^\star_\gamma + \lambda^\star(r(x) - \bar\varepsilon) \ge 0 \quad \textit{for every } x \in \mathcal{X}_\gamma. \tag{20}$$

Proof. For any feasible Borel comparator a, define $u_j = \mathbb{E}[a(S) \mid S \text{ in bin } j]$, setting it to zero on null bins, and $\mu_j = \mathbb{E}[m(S) \mid S \text{ in bin } j]$ on positive-mass bins. Conditional averaging preserves release mass. In each such bin,

$$u_j\mu_j - \mathbb{E}[a(S)m(S) \mid j] = \mathbb{E}[a(S)(\mu_j - m(S)) \mid j] \le L/K.$$

Summing with bin probabilities gives $r(u) \le \varepsilon + L/K$; also $r(u) \le 1$, so $r(u) \le \bar\varepsilon$. The vector $u^\gamma = (1-\gamma)u$ lies in $\mathcal{X}_\gamma$, remains feasible, and has $d(u^\gamma) \le c_P(a) + \gamma$. An infimizing sequence gives the first conclusion without assuming a Borel minimizer exists.

Finite-dimensional LP duality yields an optimal $\lambda^\star \ge 0$ for (19); strict feasibility follows from a sufficiently small positive vector. Its dual identity gives (20). Evaluating the dual objective at $x = 0$ gives $0 \le d^\star_\gamma \le 1 - \lambda^\star\bar\varepsilon$, hence the bound. □

Lemma 5.2 (Pathwise primal and dual regret)**.** *If $K\log(1/\gamma) \le T/4$, then, for every $x \in \mathcal{X}_\gamma$,*

$$\sum_t \langle g_t, x_t - x\rangle \le A_x := 2\sqrt{2(1+\Lambda^2)KT\log(1/\gamma)}. \tag{21}$$

For every $\lambda \in [0,\Lambda]$, without this restriction,

$$\sum_t h_t(\lambda - \lambda_t) \le A_\lambda := \frac{3}{2}\Lambda\sqrt{1 + \sum_t h_t^2}. \tag{22}$$

Proof. The key barrier cancellation is

$$q_t g_{t,b_t} = -q_t + \lambda_t D_t L_t, \qquad q_t^2 g_{t,b_t}^2 \le 2(1+\Lambda^2). \tag{23}$$

For the active coordinate, put $q = 1 - x_{t,b_t}$, $\alpha = \eta_x q g_{t,b_t}$ and $v = (x_{t+1,b_t} - x_{t,b_t})/q$. The assumed condition ensures $|\alpha| \le 1/2$. Mirror-descent optimality and the three-point Bregman identity imply

$$\begin{aligned}
\eta_x\langle g_t, x_t - x\rangle &\le D_\Psi(x,x_t) - D_\Psi(x,x_{t+1}) + \eta_x\langle g_t, x_t - x_{t+1}\rangle - D_\Psi(x_{t+1},x_t)\\
&\le D_\Psi(x,x_t) - D_\Psi(x,x_{t+1}) + \sup_{v<1}\{-\alpha v + \log(1-v) + v\}\\
&= D_\Psi(x,x_t) - D_\Psi(x,x_{t+1}) - \log(1-\alpha) - \alpha\\
&\le D_\Psi(x,x_t) - D_\Psi(x,x_{t+1}) + \alpha^2.
\end{aligned}$$

The last scalar inequality holds for $|\alpha| \le 1/2$. Since $D_\Psi(x,0) = \sum_j[-\log(1-x_j) - x_j] \le K\log(1/\gamma)$, telescoping and (23) give

$$\sum_t \langle g_t, x_t - x\rangle \le K\log(1/\gamma)/\eta_x + 2\eta_x(1+\Lambda^2)T = A_x.$$

For the dual step, nonexpansiveness of projection gives

$$h_t(\lambda - \lambda_t) \le \frac{(\lambda-\lambda_t)^2 - (\lambda - \lambda_{t+1})^2}{2\eta_{\lambda,t}} + \frac{\eta_{\lambda,t}h_t^2}{2}.$$

The step sizes decrease. Summation by parts bounds the first sum by $\Lambda\sqrt{1+B_T}/2$; the second is at most $\Lambda(\sqrt{1+B_T} - 1)$, using

$$\frac{B_t - B_{t-1}}{\sqrt{1+B_t}} \le 2(\sqrt{1+B_t} - \sqrt{1+B_{t-1}}).$$

This proves (22). $\square$

Lemma 5.3 (Martingale concentration)**.** *If* (ξ_t) *are martingale differences with* $|\xi_t| \le b$ *and* $\sum_t \mathbb{E}_{t-1}\xi_t^2 \le V$ *for deterministic* V*, then*

$$\mathbb{P}\left\{\sum_t \xi_t > \sqrt{2Vs} + bs/3\right\} \le e^{-s}, \qquad s > 0. \tag{24}$$

The same bound applies to the negative sum.

Proof. For $0 < \theta < 3/b$, expanding the conditional exponential gives

$$\mathbb{E}_{t-1}e^{\theta\xi_t} \le \exp\left\{\frac{\theta^2\mathbb{E}_{t-1}\xi_t^2}{2(1-\theta b/3)}\right\}.$$

Iteration bounds the moment-generating function of the sum by $\exp\{V\theta^2/[2(1-\theta b/3)]\}$. For $V > 0$, exponential Markov with $\theta = \sqrt{2s/V}/(1 + (b/3)\sqrt{2s/V})$ gives (24). If $V = 0$, all differences vanish almost surely. $\square$

Lemma 5.4 (A coupled population saddle bound)**.** *Suppose* $K\log(1/\gamma) \le T/4$ *and let* $x^\star$ *attain* (19)*. Write*

$$U_T = \sum_t [d(x_t) - d_\gamma^\star], \qquad V_T = \sum_t [r(x_t) - \bar{\varepsilon}].$$

On a common event of probability at least $1-\delta$*,*

$$U_T + \lambda V_T \le 25\varepsilon^{-1}H_T \quad (\lambda = 0, \Lambda), \qquad U_T \le 25\varepsilon^{-1}H_T, \quad (V_T)_+ \le 25H_T. \tag{25}$$

The event can also include

$$\sum_t D_t \le \sum_t d(x_t) + J_T, \qquad \mathcal{R}_T \le \sum_t r(x_t) + J_T, \qquad J_T = \sqrt{2T\ell_T} + \ell_T/3. \tag{26}$$

Proof. Adding (21) and (22) yields

$$\sum_t [-z_t + x_{b_t}^\star + \lambda(z_t W_t - \bar{\varepsilon})] \le A_x + A_\lambda + N_T, \qquad N_T = \sum_t \lambda_t(x_{b_t}^\star W_t - \bar{\varepsilon}). \tag{27}$$

Importance weighting and i.i.d. sampling give $\mathbb{E}_{t-1}z_t W_t = r(x_t)$ and $\mathbb{E}_{t-1}x_{b_t}^\star W_t = r(x^\star) \le \bar{\varepsilon}$. Define

$$M_0 = \sum_t [x_{b_t}^\star - z_t - \mathbb{E}_{t-1}(x_{b_t}^\star - z_t)], \qquad M_r = \sum_t [z_t W_t - r(x_t)].$$

The left side of (27) is $U_T + \lambda V_T + M_0 + \lambda M_r$. The adaptive multiplier must stay inside N_T: its nonpositive conditional drift follows from past-measurability of λ_t and feasibility of $x^\star$.

The moment calculations use $0 \le W_t \le 1/\gamma$ and $\mathbb{E}[W_t^k \mid \mathcal{F}_{t-1}, S_t] \le q_t^{1-k}$ for $k \ge 1$. They give

$$|h_t| \le 1/\gamma, \qquad \mathbb{E}_{t-1}h_t^2 \le 4/\gamma, \qquad \mathbb{E}_{t-1}h_t^4 \le 4/\gamma^3,$$
$$|z_t W_t - r(x_t)| \le 1/\gamma, \qquad \mathrm{Var}_{t-1}(z_t W_t) \le 1/\gamma.$$

For $X_t = \lambda_t(x_{b_t}^\star W_t - \bar{\varepsilon})$, $\mathbb{E}_{t-1}X_t \le 0$, $\mathbb{E}_{t-1}X_t^2 \le 4\Lambda^2/\gamma$, and $|X_t - \mathbb{E}_{t-1}X_t| \le 2\Lambda/\gamma$. Applying

Lemma 5.3, with $s = \ell_T$, gives the following simultaneous bounds:

$$
\begin{aligned}
|M_0| &\le \sqrt{2T\ell_T} + 2\ell_T/3, \\
|M_r| &\le \sqrt{2T\ell_T/\gamma} + \ell_T/(3\gamma), \\
N_T &\le \Lambda\sqrt{8T\ell_T/\gamma} + 2\Lambda\ell_T/(3\gamma), \\
\sum_t h_t^2 &\le 4T/\gamma + \sqrt{8T\ell_T/\gamma^3} + \ell_T/(3\gamma^2).
\end{aligned} \tag{28}
$$

For the last line, apply concentration to $h_t^2 - \mathbb{E}_{t-1}h_t^2$, whose range is bounded by $1/\gamma^2$ and conditional variance by $4/\gamma^3$. The two inequalities in (26) follow because $D_t, R_t \in [0,1]$ have conditional means $d(x_t), r(x_t)$. Counting the two tails for each absolute value, there are eight tails in total. Their failure probability is at most $8e^{-\ell_T} = \delta/(2KT) \le \delta$.

Put $P_T = \sqrt{KT\ell_T}$, $F_T = \sqrt{T\ell_T/\gamma}$, and $Q_T = \ell_T/\gamma$, so $H_T = P_T + F_T + Q_T$. Since $T\gamma \ge 1$ and $\ell_T \ge 1$, (28) implies $\sqrt{1 + \sum_t h_t^2} \le 5F_T + Q_T$. Also $\log(1/\gamma) \le \ell_T$. Substitution in (27), using $\Lambda = 2/\varepsilon$ and $\varepsilon^{-1} \ge 1$, bounds the coefficients of P_T, F_T, Q_T by $25\varepsilon^{-1}$ for both $\lambda = 0, \Lambda$. This proves the first inequality in (25).

Finally, summing (20) at x_t gives $U_T + \lambda^\star V_T \ge 0$. If $V_T > 0$, the case $\lambda = \Lambda$ therefore yields

$$
(\Lambda - \lambda^\star)V_T \le 25\varepsilon^{-1}H_T, \qquad \Lambda - \lambda^\star \ge 1/\varepsilon.
$$

This bounds $(V_T)_+$; the case $\lambda = 0$ bounds U_T. □

Proof of Theorem 4.1. In the stable regime $K\log(1/\gamma) \le T/4$, use the common event in Lemma 5.4. Then

$$
\mathcal{R}_T \le \bar{\varepsilon}T + 25H_T + J_T \le \varepsilon T + LT/K + G_T(\varepsilon).
$$

Similarly, by Lemma 5.1,

$$
C_T \le c_M T + T d_\gamma^\star + 25\varepsilon^{-1}H_T + J_T \le C_T^\star(\varepsilon) + \gamma T + G_T(\varepsilon).
$$

The coefficient $100(1+\varepsilon^{-1})^2$ bounds all the displayed remainders. Both sides include the same $c_M T$; neither the algorithm nor an optimal comparator incurs audit correction charges.

If $K\log(1/\gamma) > T/4$, then $\sqrt{KT\ell_T} > T/2$ and $G_T(\varepsilon) \ge T$. The deterministic bounds $\mathcal{R}_T \le T$ and $C_T - C_T^\star(\varepsilon) \le T$ establish (15). This is a proof case; the scalar update (12) remains well-defined and the algorithm does not switch policies. □

6 A complementary lower bound

We now specialize the same model to binary labels, the constant predictor $\widehat{Y}_t = 0$, costs $c_M = c_R = 0$, risk budget $\varepsilon = 1/4$, and Lipschitz bound $L = 1$. Thus $L_t = Y_t$, $C_T = \sum_t I_t$, and $\mathcal{R}_T = \sum_t (1 - I_t)Y_t$. The lower bound allows every adaptive release/defer/audit learner, not only binned policies or OBSR. Its initial state and randomness cannot depend on the unknown instance. For an instance P_θ, abbreviate the same population benchmark by

$$
C_{\theta,T}^\star = T \inf_{a:\, \mathbb{E}_\theta[a(S)m_\theta(S)] \le 1/4} \mathbb{E}_\theta[1 - a(S)]. \tag{29}
$$

The infimum is again over all Borel $a : [0,1] \to [0,1]$. The following construction is called CICAT (Camouflaged Independent-Cell Apple Tasting).

Theorem 6.1 (CICAT cost–error lower bound). *For every integer $T \geq 8$, there is a finite family $\{P_\theta : \theta \in \Theta_T\}$ satisfying the specialized model in Section 6, with $\Theta_T = \{-1, +1\}^{\lfloor T^{1/3} \rfloor}$, such that every admissible randomized adaptive learner satisfies*

$$\max_{\theta \in \Theta_T} \left\{ \mathbb{E}_\theta C_T - C^\star_{\theta,T} + 2\left(\mathbb{E}_\theta \mathcal{R}_T - \frac{T}{4} \right) \right\} \geq \frac{T^{2/3}}{2048}. \tag{30}$$

Here $C^\star_{\theta,T} = C^\star_{P_\theta,T}(1/4)$, and $\mathbb{E}_\theta$ includes the arrivals, labels, and learner randomization. The learner may know T, the score marginal, and the entire family, but not the unknown θ. Every instance has an optimal comparator with risk exactly $1/4$.

Corollary 6.2 (Cost lower bound under controlled error slack). *If a learner satisfies*

$$\mathbb{E}_\theta \mathcal{R}_T \leq \frac{T}{4} + \frac{T^{2/3}}{8192} \qquad \textit{for every } \theta \in \Theta_T, \tag{31}$$

then, on at least one instance,

$$\mathbb{E}_\theta C_T - C^\star_{\theta,T} \geq \frac{T^{2/3}}{4096}. \tag{32}$$

Proof. Choose an instance attaining the maximum in (30) and subtract twice the error excess. Equations (30)–(31) give $\mathbb{E}_\theta C_T - C^\star_{\theta,T} \geq (1/2048 - 2/8192)T^{2/3} = T^{2/3}/4096$. ☐

The proof in Appendix A constructs the family, identifies its binding optimal comparator, and bounds the information available to any adaptive learner. The signed error excess in (30) is essential: the theorem alone is not a cost lower bound for a learner allowed arbitrary error slack.

6.1 A precise optimality consequence for nominal PAC labeling

Target tightening permits a common comparison criterion for the upper and lower results. The following definition fixes both the error requirement and the comparator before comparing their rates. Let $\mathcal{P}_1$ be the fixed class of all i.i.d. laws of $(S, Y) \in [0,1] \times \{0,1\}$ with a globally one-Lipschitz version of $\mathbb{P}(Y = 1 \mid S = s)$. Fix $\widehat{Y} \equiv 0$, $c_M = c_R = 0$, unit expert cost, and $\varepsilon = 1/4$. For horizon T, define

$$\begin{aligned} \mathcal{A}_T^{\mathrm{PAC}} &= \left\{ \mathcal{A} : \sup_{P \in \mathcal{P}_1} \mathbb{P}_P(\mathcal{R}_T > T/4) \leq T^{-2} \right\}, \\ \mathfrak{M}_T &= \inf_{\mathcal{A} \in \mathcal{A}_T^{\mathrm{PAC}}} \sup_{P \in \mathcal{P}_1} \{\mathbb{E}_P C_T - C^\star_{P,T}(1/4)\}. \end{aligned} \tag{33}$$

The learners obey the information protocol and are fixed independently of the unknown P; they may know the class and horizon. The class is nonempty because always using the expert is admissible.

Corollary 6.3 (Minimax exponent in a nominal-PAC class). *There are universal constants $C < \infty$ and T_0 such that, for all integers $T \geq T_0$,*

$$\frac{T^{2/3}}{4096} \leq \mathfrak{M}_T \leq C T^{2/3} \sqrt{\log T}. \tag{34}$$

Proof. For any $\mathcal{A} \in \mathcal{A}_T^{\mathrm{PAC}}$, bounded loss gives $\mathbb{E}_P \mathcal{R}_T \leq T/4 + T \cdot T^{-2} = T/4 + T^{-1}$, uniformly in P. The finite CICAT family is contained in the fixed class $\mathcal{P}_1$. For all sufficiently large T, $T^{-1} \leq T^{2/3}/8192$, so Corollary 6.2 gives the lower bound for each admissible learner and hence for their infimum.

For the upper bound, take the tightened OBSR procedure in Corollary 4.3, with $\varepsilon = 1/4$, $L = 1$, $\delta = T^{-2}$, $K = \lceil T^{1/3} \rceil$, and $\gamma = T^{-1/3}$. Its finite-sample condition holds for all sufficiently large T, uniformly over $\mathcal{P}_1$. It therefore belongs to $\mathcal{A}_T^{\mathrm{PAC}}$ and has a uniform high-probability cost gap $O(T^{2/3}\sqrt{\log T})$. On the failure event, $C_T - C^{\star}_{P,T}(1/4) \le T$; taking expectations adds at most T^{-1}. Taking the supremum over P establishes the upper bound. □

This establishes the exponent for the specific criterion (33), up to a square-root logarithmic factor. It does not characterize every error–cost tradeoff or imply the same cost lower bound when failure probability is fixed, error slack is larger, or the score regularity and comparator class change. In particular, a fixed failure probability would contribute $O(T)$ in the expectation conversion used above.

7 Scope and empirical evaluation

Practical scope of the theory. OBSR learns a routing policy for a fixed predictor and score. Its bounds apply to realized deployment costs and released errors, with expert queries supplying all online label feedback. They do not provide a guarantee for an adaptive pipeline that also retrains the predictor or score on those labels. The nominal-PAC and minimax conclusions depend on the explicit tightening condition and expectation conversion above. The theorem constants may be too conservative to justify savings at available dataset sizes. Sharper finite-sample analysis remains valuable even if the asymptotic exponent is settled for this model.

7.1 A fully specified calibration baseline

For a population-stream comparison, consider a calibrated threshold baseline inspired by PAC labeling [1]. The following independent-sample construction has its own deployment certificate; it is not the published fixed-pool algorithm. Fix the predictor and score before drawing $n \ge 1$ calibration pairs (S_i', L_i') i.i.d. from the same law of (S, L) as the deployment stream, independently of that stream. For $j = 0, \dots, 200$, let $a_j(s) = \mathbf{1}\{s \le j/200\}$ and define

$$\widehat{r}_j = \frac{1}{n}\sum_{i=1}^{n} a_j(S_i')L_i', \qquad u_n = \sqrt{\frac{\log(402/\delta)}{2n}}, \qquad v_T = \sqrt{\frac{\log(2/\delta)}{2T}}. \tag{35}$$

Choose the largest j with $\widehat{r}_j + u_n + v_T \le \varepsilon$. If none exists, use the expert-only policy $a \equiv 0$, whose risk is known to be zero. Freeze the selected policy and deploy it on an independent stream of T items. The largest feasible threshold minimizes expert usage among the certified thresholds because their release sets are nested, regardless of whether m is monotone.

For completeness, Hoeffding's inequality and a union bound over 201 thresholds give $r_P(a_j) \le \widehat{r}_j + u_n$ for every j with probability at least $1 - \delta/2$. Conditional on calibration, a second Hoeffding bound gives deployment average released error at most $r_P(a_j) + v_T$ with probability at least $1 - \delta/2$. Thus this baseline satisfies $\mathbb{P}(\mathcal{R}_T \le \varepsilon T) \ge 1 - \delta$; the expert-only fallback is safe deterministically. This certificate controls error, but supplies no regret guarantee against all Borel score policies. Its threshold restriction can exclude useful policies when m is nonmonotone. Charge $n(1 + c_M)$ for calibration in addition to deployment cost; report both quantities. An end-to-end comparison must also disclose any labels used to train the fixed predictor or score.

7.2 Proposed validation protocol

No empirical OBSR result is claimed in this draft. A first comparison should include balanced OBSR, its tightened variant when the sufficient condition holds, the baseline in (35), and always-expert labeling. The distribution-aware score-policy optimum is an evaluation-only oracle. A separate batch study can compare the published PAC-labeling procedure [1] under its fixed-pool protocol. Report the difference in pool access and use a common ledger for all incurred model calls and expert queries; its finite-pool certificate must be interpreted in that sampling model.

For a synthetic correctness study, draw i.i.d. scores $S_t \sim \mathrm{Uniform}[0,1]$, then conditionally independently draw $Y_t \mid S_t \sim \mathrm{Ber}(m(S_t))$, with $\widehat{Y}_t = 0$. Use $m(s) = 0.05 + 0.4s$ and $m(s) = 0.05 + 0.4(2s-1)^2$, whose Lipschitz bounds are 0.4 and 1.6. These are synthetic mechanisms with known oracle risks, not real model evidence. A reproducible initial sweep would use $T \in \{2^{10}, 2^{12}, 2^{14}, 2^{16}, 2^{18}\}$, $\varepsilon \in \{0.05, 0.10, 0.15\}$, $\delta = 0.05$, and 500 trials with seed $20260908 + r$ for trial $r = 0, \ldots, 499$; methods share each trial's stream and use separate routing random-number streams. Use $n = \lceil 0.2T \rceil$ for the independent calibration baseline and the threshold grid and confidence allocation in (35). This fixed calibration fraction is a benchmark choice, not a sublinear-cost claim. Use $c_M \in \{0, 0.05\}$, $c_R = 0$, and unit expert cost. OBSR uses the displayed balanced parameters without oracle tuning. Record whether the target-tightening condition holds at each horizon; if it fails, report that fact and the safe expert-only fallback separately from the tightened method's cost bound.

Report actual released error, latent counterfactual model error used only for evaluation, expert-query frequency, all cost components, and gap to the same oracle comparator. Report violations both at nominal ε and at the theorem's stated finite-horizon bound. For a real benchmark, use independent held-out examples from genuine model prediction tables. Repeated permutations of a fixed dataset do not reproduce i.i.d. sampling conditional on that dataset; distinguish such finite-dataset stress tests from the theorem's sampling model. Distribution-shift studies must likewise be labeled outside the proven stationary setting. A multi-source extension requires genuine distinct models and a new theorem for its observation and cost structure.

Remaining research obligations. Independent proof review, a broader novelty audit against the public literature, and an implementation with reproducible empirical validation remain necessary before submission. Investigate whether confidence-adaptive querying or sharper concentration improves finite-sample cost while retaining nominal error control. Unknown Lipschitz bounds, changing scores, multiple costly sources, and imperfect experts require additional analysis; they are not covered by the results above.

A Proof of the CICAT lower bound

A.1 The finite family

Fix $T \geq 8$ and set

$$M = \lfloor T^{1/3} \rfloor, \qquad \beta = \frac{1}{64(M+1)}, \qquad \bar{\theta} = \frac{1}{M}\sum_{i=1}^{M} \theta_i, \qquad s_i = \frac{1}{2} + \frac{i}{2(M+1)}. \tag{36}$$

Here β is a signal gap, not a failure probability. We have $M \geq 2$ and $\beta \leq 1/192$. The score marginal is the same for all $\theta \in \{-1, +1\}^M$, and the conditional law is $Y \mid S = s \sim \mathrm{Ber}(m_\theta(s))$, with

Score	Probability	Conditional error $m_\theta(s)$
0 (anchor)	1/4	$1/8 - \beta\bar{\theta}$
1/2 (boundary)	1/2	1/2
s_i, $i = 1, \ldots, M$	$1/(4M)$	$1/2 + \theta_i\beta$

Between consecutive support points, extend m_θ by linear interpolation, and keep it constant on $[s_M, 1]$. Draw the T pairs independently from this law. This is a constructed lower-bound family, not an empirical dataset.

Lemma A.1 (Admissibility and common marginals)**.** *Every P_θ satisfies (2) with $L = 1$. The score marginal, the model output, and the marginal label law are identical across the family; in particular, $\mathbb{E}_\theta Y = 13/32$.*

Proof. The probabilities sum to one, and all conditional means lie in $(0, 1)$. The interpolation slope on $[0, 1/2]$ is $3/4 + 2\beta\bar{\theta}$, whose absolute value is less than one. On $[1/2, s_1]$ its absolute value is $2\beta(M + 1) = 1/32$; between successive s_i it is at most $4\beta(M + 1) = 1/16$; after s_M it is zero. Hence the continuous extension is globally one-Lipschitz. Finally,

$$\mathbb{E}_\theta Y = \frac{1}{4}\left(\frac{1}{8} - \beta\bar{\theta}\right) + \frac{1}{4M}\sum_{i=1}^{M}\left(\frac{1}{2} + \theta_i\beta\right) + \frac{1}{2}\cdot\frac{1}{2} = \frac{13}{32}.$$

Since Y is binary, its marginal law is therefore common as well. $\square$

A.2 Optimal comparator and the cost–error decomposition

Let $n_- = \#\{i : \theta_i = -1\} = M(1 - \bar{\theta})/2$ and define

$$\begin{aligned} r_\theta &= \frac{1}{4}\left(\frac{1}{8} - \beta\bar{\theta}\right) + \frac{n_-}{4M}\left(\frac{1}{2} - \beta\right) \\ &= \frac{3}{32} - \frac{\beta}{8} - \bar{\theta}\left(\frac{1}{16} + \frac{\beta}{8}\right), \qquad \alpha_\theta = 1 - 4r_\theta. \end{aligned} \tag{37}$$

Define $a^\star_\theta$ on the support by

$$a^\star_\theta(0) = 1, \qquad a^\star_\theta(1/2) = \alpha_\theta, \qquad a^\star_\theta(s_i) = \mathbf{1}\{\theta_i = -1\}, \tag{38}$$

and extend it as a Borel function, for example by setting it to zero off the support.

Lemma A.2 (Optimality and excess objective)**.** *The policy $a^\star_\theta$ is optimal in (29), has risk exactly $1/4$, and satisfies*

$$C^\star_{\theta,T} = T\left(\frac{5}{16} - \frac{\beta}{4}(1 + \bar{\theta})\right). \tag{39}$$

For any learner, set

$$\Delta_\theta = \mathbb{E}_\theta C_T - C^\star_{\theta,T} + 2(\mathbb{E}_\theta \mathcal{R}_T - T/4), \tag{40}$$

$$W_i = \sum_{t=1}^{T}\mathbf{1}\{S_t = s_i\}(\mathbf{1}\{\theta_i = -1\}I_t + \mathbf{1}\{\theta_i = +1\}(1 - I_t)). \tag{41}$$

Then

$$\Delta_\theta \geq 2\beta\sum_{i=1}^{M}\mathbb{E}_\theta W_i. \tag{42}$$

Proof. Equation (37) gives $1/32 - \beta/4 \le r_\theta \le 5/32$, so $0 < r_\theta < 1/4$ and $0 < \alpha_\theta < 1$. The anchor and negative cells contribute risk r_θ, and the boundary contributes $\alpha_\theta/4$. Thus $r_{P_\theta}(a^\star_\theta) = 1/4$. Its deferral probability is $(M - n_-)/(4M) + (1-\alpha_\theta)/2$; simplifying yields (39) once optimality is established.

For a policy a, introduce

$$J_\theta(a) = c_{P_\theta}(a) + 2r_{P_\theta}(a) = \mathbb{E}_\theta[1 + a(S)(2m_\theta(S) - 1)].$$

The policy (38) minimizes the integrand at every support point: it releases below error $1/2$, reveals above $1/2$, and may mix at equality. For any feasible a,

$$c_{P_\theta}(a) = J_\theta(a) - 2r_{P_\theta}(a) \ge J_\theta(a^\star_\theta) - \frac{1}{2} = c_{P_\theta}(a^\star_\theta),$$

proving optimality.

Because the action precedes the label and the stream is i.i.d.,

$$\mathbb{E}_\theta[I_t + 2R_t \mid \mathcal{H}_{t-1}, S_t, A_t] = I_t + 2(1 - I_t)m_\theta(S_t).$$

The comparator's expected total objective is $TJ_\theta(a^\star_\theta) = C^\star_{\theta,T} + T/2$. Hence Δ_θ is the learner's excess expected objective over this pointwise minimum. At cell i, revealing when $\theta_i = -1$ or releasing when $\theta_i = +1$ incurs excess 2β. The excess is zero at the boundary and nonnegative at the anchor. Summing and discarding the anchor excess gives (42). $\square$

A.3 Indistinguishability and adaptive decisions

For probability laws P, Q, write $D_{\mathrm{KL}}(P\|Q)$ for relative entropy with natural logarithms, and $\mathrm{TV}(P,Q) = \sup_B |P(B) - Q(B)|$. Fix a coordinate i and all other signs. Denote the two resulting instances by θ^- and θ^+, according to the sign of coordinate i.

Lemma A.3 (Information bound)**.** *For every fixed learner and every $t \le T$, the laws ν_t^- and ν_t^+ of $\mathcal{H}_{t-1}$ under the paired instances satisfy*

$$D_{\mathrm{KL}}(\nu_t^- \| \nu_t^+) < \frac{1}{100}, \qquad \mathrm{TV}(\nu_t^-, \nu_t^+) < \frac{1}{4}. \tag{43}$$

The same bounds hold for the complete observable interaction.

Proof. For $p, q \in (0,1)$, $\log x \le x - 1$ implies

$$D_{\mathrm{KL}}(\mathrm{Ber}(p) \| \mathrm{Ber}(q)) \le \frac{(p-q)^2}{q(1-q)}. \tag{44}$$

The paired laws differ only at s_i and the anchor. At s_i their means differ by 2β, and $q(1-q) = 1/4 - \beta^2 > 4/17$, so the conditional divergence is at most $17\beta^2$. At the anchor the difference is $2\beta/M$, and

$$q \in [23/192, 25/192], \qquad q(1-q) \ge \frac{23 \cdot 169}{192^2} > \frac{1}{10}.$$

Its conditional divergence is therefore at most $40\beta^2/M^2$.

Consider the full latent sample $(S_t, Y_t)_{t=1}^T$, including labels never revealed to the learner, and denote its law by Q_θ. The common score marginal and product additivity give

$$\begin{aligned} D_{\mathrm{KL}}(Q_{\theta^-}\|Q_{\theta^+}) &\le T\left(\frac{17\beta^2}{4M}+\frac{10\beta^2}{M^2}\right) \\ &\le T\beta^2\left(\frac{17}{4M}+\frac{40}{M^2}\right) \\ &< \frac{M+1}{4096}\left(\frac{17}{4M}+\frac{40}{M^2}\right) \le \frac{291}{32768} < \frac{1}{100}. \end{aligned} \tag{45}$$

Here $T<(M+1)^3$, and, for $M\ge 2$,

$$(M+1)\left(\frac{17}{4M}+\frac{40}{M^2}\right)=\frac{17}{4}+\frac{177}{4M}+\frac{40}{M^2}\le\frac{291}{8}.$$

The learner's random seed is independent of this sample and has the same law under both instances. Every observable history, and the entire interaction, is a common randomized transformation of the latent sample. Data processing transfers (45) to these laws. Pinsker's inequality then gives $\mathrm{TV}\le\sqrt{D_{\mathrm{KL}}/2}<1/\sqrt{200}<1/4$. □

Lemma A.4 (Paired action errors)**.** *For every paired θ^-,θ^+,*

$$\mathbb{E}_{\theta^-}\sum_{t=1}^{T}\mathbf{1}\{S_t=s_i\}I_t+\mathbb{E}_{\theta^+}\sum_{t=1}^{T}\mathbf{1}\{S_t=s_i\}(1-I_t)\ge\frac{3T}{16M}. \tag{46}$$

Proof. Let $k_{t,i}(h)\in[0,1]$ be the learner's probability of revealing at score s_i given history h, integrating over its fresh randomness. This is a common kernel under the two instances. Put $q_t^{\pm}=\int k_{t,i}\,d\nu_t^{\pm}$. By Lemma A.3,

$$|q_t^- - q_t^+|\le \mathrm{TV}(\nu_t^-,\nu_t^+)<\frac{1}{4},\qquad q_t^-+1-q_t^+\ge\frac{3}{4}.$$

Under each instance, S_t is independent of the pre-score history and $\mathbb{P}(S_t=s_i)=1/(4M)$. Thus the left side of (46) equals

$$\frac{1}{4M}\sum_{t=1}^{T}(q_t^-+1-q_t^+)\ge\frac{3T}{16M}.$$

Comparing histories *before* the current score arrives is essential: no conditioning of a joint total-variation bound on a rare score event is used. □

A.4 Completion of the lower-bound proof

Proof of Theorem 6.1. Lemma A.1 establishes admissibility, and Lemma A.2 supplies the binding optimal comparator. For each coordinate i, partition Θ_T into the 2^{M-1} pairs differing only at i. Lemma A.4 gives

$$\frac{1}{2^M}\sum_{\theta\in\Theta_T}\mathbb{E}_\theta W_i\ge\frac{2^{M-1}}{2^M}\frac{3T}{16M}=\frac{3T}{32M}.$$

Using (42) and summing over i,

$$\max_\theta\Delta_\theta\ge\frac{1}{2^M}\sum_\theta\Delta_\theta\ge 2\beta\frac{3T}{32}=\frac{3T}{1024(M+1)}.$$

Since $T^{1/3}\ge 2$ and $M+1\le T^{1/3}+1\le(3/2)T^{1/3}$, the last quantity is at least $T^{2/3}/512$, which is stronger than the stated $T^{2/3}/2048$. This proves (30). □

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